Fahim Tasneema Azad

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Research Interests

My research focuses on developing scalable, causally-informed machine learning methods for large-scale, heterogeneous multivariate time series, where data are high-dimensional, noisy, sparse, and subject to regime shift. My work begins with a deep understanding of the data, such as its structure, causal dependencies, and sources of shift and sparsity, uses that understanding to develop efficient and robust learning algorithms, and validates them on real-world data from high-impact complex dynamic systems. Over 12 years of data-centric research and practice, I have worked at the intersection of computing and application domains, collaborating closely with experts in industry and enterprise systems, public health, and sustainability to bridge domain knowledge and data-driven modeling, and to turn large, messy real-world data into reliable forecasts and actionable decisions.

Education

2026 (Expected) **Ph.D. in Computer Science**, *Arizona State University*, Tempe, United States
Advisor: Prof. K. Selçuk Candan. GPA: 4.14/4.00.

2023 **M.S. in Computer Science**, *Arizona State University*, Tempe, United States
GPA: 4.14/4.00 (with Distinction).

2014 **M.B.A. in Finance**, *Institute of Business Administration, University of Dhaka*, Dhaka, Bangladesh

2009 **B.IT. in Software Engineering**, *Multimedia University*, Cyberjaya, Malaysia
GPA: 3.71/4.00 (with Distinction).

Research Experience

Aug 2019 – Present **Graduate Research Assistant**, *EMIT Lab, Arizona State University*, Tempe, United States
Advisor: Prof. K. Selçuk Candan.

- Designed a factorization-machine-based attention mechanism that leverages second-order feature interactions to reduce self-attention computation in transformer-based multivariate time series forecasting; addresses the quadratic-complexity bottleneck for sparse temporal data.
- Developed causally-regularized representation learning for multivariate time series: injection of directed lagged causal structure into factorization-machine interaction matrices via (i) interaction masking, (ii) Frobenius-norm regularization on $VU^\top$, and (iii) a hybrid of the two, improving forecasting accuracy under regime shift.
- Proposed a domain disentanglement framework for transferable spatio-temporal repre-

sentations, enabling selective knowledge transfer from data-rich to data-poor settings without target supervision.
- Developed multivariate spatio-causal forecasting methods that combine mechanistic priors with learned representations for complex dynamic systems under data sparsity and distribution shift.
- Built a simulation-ensemble framework for counterfactual reasoning over complex coupled dynamic systems, supporting exploration of alternative scenario timelines and intervention analysis.
- Investigated LLM-driven hypothesis generation conditioned on spatio-causal structure and mechanistic knowledge for scientific discovery in temporal data.

Jun 2026 – Aug 2026 **Summer Research Intern**, *NSF Analysis and Predictions of Pandemic Expansion (APPEX) Center*, Tempe, United States

- Developed an agentic-AI framework for rule-based causal discovery from published literature, orchestrating LLM-based agents to propose, test, and refine candidate causal structures for use in downstream predictive hypothesis generation.

Jun 2007 – Sep 2007 **Research Intern**, *Network Research Group, School of Computer Sciences, Universiti Sains Malaysia*, Penang, Malaysia
Advisor: Dr. Sureswaran Ramadass.

- Implemented an interface for Unidirectional Lightweight Encapsulation with header compression for IP-based satellite communication over DVB-S ("Administration Tools for UDL Gateway").

Professional Appointments

Jan 2015 – Jul 2018 **Assistant Professor** (2016–2018)**, Senior Lecturer** (2016)**, Lecturer** (2015–2016), *United International University*, Dhaka, Bangladesh

- Sole instructor for undergraduate-level Database Management Systems, Programming Languages, and E-Commerce; designed curriculum, lectured, and evaluated 80 students per course across multiple semesters.
- Led the faculty team preparing departmental documentation for ACBSP accreditation, including the on-site presentation.
- Secured a research grant as Principal Investigator (BDT 5,10,000) and supervised undergraduate research interns.

Oct 2010 – May 2012 **Lecturer**, *State University of Bangladesh*, Dhaka, Bangladesh

- Sole instructor for undergraduate Data Structures, Discrete Mathematics, and Digital Logic Design across multiple cohorts of 80 students.

Publications

Peer-reviewed Journal and Conference Articles

[J1] **Fahim Tasneema Azad**, K. Selçuk Candan, Ahmet Kapkiç, Mao-Lin Li, Huan Liu, Pratanu Mandal, Paras Seth, Bilgehan Arslan, Gerardo Chowell-Puente, John Sabo, Rebecca Muenich, Javier Redondo, and Maria Luisa Sapino, "A Vision for Spatio-Causal Situation Awareness, Forecasting, and Planning," *ACM Transactions on Spatial Algorithms and Systems (TSAS)*, 2024.

[J2] Nina H. Fefferman, John S. McAlister, Belinda S. Akpa, Kelechi Akwataghibe, **Fahim Tasneema Azad**, K. Selçuk Candan, Gerardo Chowell-Puente, et al., "A New Paradigm for Pandemic Preparedness," *Current Epidemiology Reports*, Springer, 2023.

[J3] **Fahim Tasneema Azad**, Robert W. Dodge, Allen M. Varghese, Jaejin Lee, Giulia Pedrielli, K. Selçuk Candan, and Gerardo Chowell-Puente, "SIRTEM: Spatially Informed Rapid Testing for Epidemic Modeling and Response," *ACM Transactions on Spatial Algorithms and Systems (TSAS)*, 2022.

[C1] Yoonhyuk Choi and **Fahim Tasneema Azad**, "Improving the Text Convolution Mechanism with Large Language Models for Review-Based Recommendation," in *IEEE International Conference on Big Data (Big Data)*, 2024.

Peer-reviewed Workshop Papers and Posters

[W1] **Fahim Tasneema Azad**, "Factorization-Machine-Enhanced Crossformer for Multivariate Time Series Event Detection," in *Women in Machine Learning (WiML) Workshop at NeurIPS*, 2025.

[W2] **Fahim Tasneema Azad** and K. Selçuk Candan, "FMAM: A Factorization-Machine-Based Attention Mechanism for Time Series Forecasting," in *Women in Machine Learning (WiML) Workshop at NeurIPS*, 2022.

[P1] **Fahim Tasneema Azad**, K. Selçuk Candan, and Gerardo Chowell-Puente, "Domain Disentanglement for Epidemic Onset Forecasting: Knowledge Transfer from Data-Rich to Data-Poor Communities," in *10th International Conference on Infectious Disease Dynamics (Epidemics10)*, 2025.

[P2] **Fahim Tasneema Azad**, Javier Redondo Anton, Shubhodeep Mitra, Fateh Singh, Hans Walter Behrens, Mao-Lin Li, Bilgehan Arslan, K. Selçuk Candan, Maria Luisa Sapino, Gerardo Chowell-Puente, and Giulia Pedrielli, "PanCommunity: Non-Monolithic Complex Epidemic and Pandemic Modeling," in *9th International Conference on Infectious Disease Dynamics (Epidemics9)*, 2023.

[P3] **Fahim Tasneema Azad**, Robert W. Dodge, Allen M. Varghese, Jaejin Lee, Giulia Pedrielli, K. Selçuk Candan, and Gerardo Chowell-Puente, "SIRTEM: Spatially Informed Rapid Testing for Epidemic Modeling and Response to COVID-19," in *9th International Conference on Infectious Disease Dynamics (Epidemics9)*, 2023.

Peer-reviewed Tutorials

[TU1] **Fahim Tasneema Azad**, Ahmet Kapkiç, Pratanu Mandal, Abhinav Gorantla, Shu Wan, Maria Luisa Sapino, Huan Liu, and K. Selçuk Candan, "Spatio-Causal Modeling and Applications," in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*, 2026. To appear.

White Papers

[WP1] **Fahim Tasneema Azad**, Javier Redondo Anton, Shubhodeep Mitra, Fateh Singh, Hans Behrens, Mao-Lin Li, Bilgehan Arslan, K. Selçuk Candan, and Maria Luisa

Sapino, "DataStorm-EM: Exploration of Alternative Timelines within Continuous-Coupled Simulation Ensembles," *arXiv preprint arXiv:2407.14571*, 2024.

Selected Research Projects

Jan 2024 – Present **Causally-Informed Factorization Machine for Multivariate Time Series Forecasting**
EMIT Lab, Arizona State University; Advisor: Prof. K. Selçuk Candan

- Generalized factorization-machine formulation with lag-augmented inputs and asymmetric low-rank interactions $VU^\top$ for directed lagged causal effects.
- Three causal-injection strategies (interaction mask, Frobenius regularization, hybrid) compared against non-causal baselines; analysis of how graph density and direction asymmetry affect forecasting error.

Jan 2022 – Present **Domain Disentanglement for Transferable Spatio-Temporal Representations**
EMIT Lab, Arizona State University; Advisor: Prof. K. Selçuk Candan

- Disentangled representation framework that separates domain-invariant from domain-specific structure to support selective knowledge transfer from data-rich to data-poor settings.
- Validated on epidemic onset forecasting across multiple communities under regime shift.

Jan 2022 – 2024 **Simulation Ensembles for Counterfactual Reasoning in Coupled Dynamic Systems**
EMIT Lab, Arizona State University; Advisor: Prof. K. Selçuk Candan

- Non-monolithic simulation-ensemble framework (PanCommunity / DataStorm-EM) enabling exploration of alternative timelines and intervention analysis in complex coupled systems.

Jan 2022 – May 2024 **Factorization-Machine Attention for Efficient Transformer Forecasting**
EMIT Lab, Arizona State University; Advisor: Prof. K. Selçuk Candan

- Survey of transformer-based time-series forecasting; design of an attention mechanism exploiting second-order feature interactions to reduce self-attention computation cost.

May 2020 – Dec 2021 **Spatially-Informed Optimization for Epidemic Resource Allocation**
EMIT Lab, Arizona State University; Advisor: Prof. K. Selçuk Candan

- Single-city and multi-city spatio-temporal forecasting integrated with constrained optimization for allocation of limited testing resources.

Funding and Grant Proposals

2025 **National Science Foundation**. Contributed (with advisor Prof. K. Selçuk Candan) to the successful NSF proposal *PanAX: Accelerating Epidemic Science through Causally-Informed Domain Generalization and Knowledge Transfer*, funded in 2025.

2025 **Arizona State University**. AI Innovation Challenge Grant (computing resources for research) for Summer 2025.

2016 **United International University Research Grant Program**. Principal Investigator, *Investigation of Critical Failure Factors Affecting ERP Implementation in the Textile and Apparel Industry in Bangladesh* (**BDT 5,10,000** $\simeq$ **USD 4147**).

Academic Service

Conference Program Committee

2026 Unifying Representations in Neural Models (UniReps Neurips)
2026 ACM Multimedia (ACMMM)
2026 IEEE Big Data
2026 Trustworthy Agentic Systems (TAS) AAAI-26 Fall Symposium Series

Conference & Workshop Reviewer

2027 AAAI Undergraduate Consortium (AAAI-UC)
2026 Neurips Workshop FMTS
2026 ICLR Workshop TSALM
2022, 2026 KDD
2026 DSAA
2025, 2026 WiML at NeurIPS
2024 CIKM
2025 ACM Multimedia (ACMMM)

Journal Reviewer

2025, 2026 ACM Computing Surveys
2026 IEEE TKDD
2021 IEEE TKDE
2025, 2026 IEEE Multimedia
2026 ACM AI Letters

Conference Volunteer

2022, 2025 WiML, NeurIPS
2022 WSDM
2020 SIGMOD

Invited Talks

2023 *Causally-Informed Multivariate Time Series Forecasting in Complex Systems*, **CRA-WP Grad Cohort for Women**.
2023 *Causally-Informed Multivariate Time Series Forecasting in Complex Coupled Systems*, **University of Torino, Italy**.
2023 *Causally-Informed Multivariate Time Series Forecasting in Complex Coupled Systems*, **Eurecom, France**.

Teaching

Arizona State University

Fall 2025 **Instructor of Record**, Knowledge Representation and Reasoning (CSE 579, graduate); 350 graduate students. Delivered lectures; prepared lecture materials.

Fall 2025 **Teaching Assistant**, Multimedia and Web Databases (CSE 515, graduate). Led project demonstrations, designed rubrics, evaluated reports.

Fall 2019 – Spring 2020 **Teaching Assistant**, Introduction to Programming Languages (CSE 240); 120 undergraduates. Taught C++ concepts, designed materials, evaluated students.

Spring 2020 **Teaching Assistant**, Principles of Programming (Java) (CSE 110); 120 undergraduates. Taught Java concepts and graded.

Spring 2019 **Teaching Assistant**, Software Analysis and Design (CSE 460). Designed weekly assignments and exams.

Spring 2019 **Teaching Assistant**, Object-Oriented Programming and Data Structures (CSE 205); 120 undergraduates. Taught Java concepts and graded.

Multimedia University, Malaysia

Summer 2008 **Teaching Assistant**, Principles of Programming (C++) (CSE 110).

Honors & Awards

Fellowships, Scholarships & Awards

2025, 2026 **Graduate Student Government, ASU**. Outstanding Mentorship Award (Spring 2025 and Spring 2026).

Spring 2025 **Ira A. Fulton Schools of Engineering, ASU**. Engineering Graduate Fellowship.

2024–2025 **Ira A. Fulton Schools of Engineering**. Harold & Lucille Dunn Memorial Engineering Scholarship.

Spring 2024 **School of Computing and Augmented Intelligence, ASU**. SCAI Doctoral Fellowship.

2023–2024 **School of Computing and Augmented Intelligence, ASU**. SCAI Outstanding CS M.S. Thesis Award.

2023–2024 **Pitchfork Awards, ASU**. Outstanding Graduate Student Leader Award.

2023 **Arizona State University**. ASU Impact Award, for excellence in research, leadership, and service.

2022–2025 **Travel Awards & Conference Scholarships**. NeurIPS / WiML Travel Awards (2022 and 2025); AnitaB.org GHC Scholarships (2022 and 2023); CRA-WP Grad Cohort Scholarship (2023); SCAI Travel Fellowship, NeurIPS (2022); SWE Conference Scholarship (2022).

June 2010 **Advanced Digital Sciences Center (ADSC) and University of Illinois Urbana-Champaign (UIUC) collaboration: Agency for Science, Technology and Research (A*STAR), Singapore**. Scholarship recipient, High Performance Computing Course & Symposium; course conducted by Dr. William Gropp, Dr. Marc Snir, Dr. Josep Torrellas, and Dr. Laxmikant Kale.

2010 **Multimedia University**. Book Award for excellence in undergraduate studies.

2005–2009 **Multimedia University**. Dean's List Awards (undergraduate).

2005–2009 **Telecom Malaysia Scholarship** to pursue undergraduate studies.

Honor Societies

Tau Beta Pi, Engineering Honor Society; invited and elected to the Arizona Beta Chapter based on exemplary character and academic standing in the top one-eighth of engineering juniors or top one-fifth of seniors.

Sigma Xi, The Scientific Research Honor Society; nominated and elected to Full Membership in recognition of noteworthy achievement as an original researcher.

Upsilon Pi Epsilon, International Honor Society for the Computing and Information Disciplines; elected based on academic distinction in computing; President, Arizona State University Chapter.

IEEE–Eta Kappa Nu (IEEE-HKN), Electrical and Computer Engineering Honor Society; elected to the Epsilon Beta Chapter based on academic distinction and professional character.

The National Society of Leadership and Success (NSLS); invited and inducted into the Arizona State University Chapter as one of fewer than 5% of students nominated, based on academic achievement and demonstrated leadership potential.

Golden Key International Honour Society; invited based on high academic standing; membership is limited to the top 15% of eligible undergraduates and top-performing graduate students; Member and Chapter Officer.

Industry Experience

Feb 2014 – Feb 2015 **Management Trainee Officer**, *United Finance Ltd.*, Dhaka, Bangladesh

Led a 14-developer team to design, implement, and deploy an end-to-end lease-approval system used across multiple branches; trained company-wide users on the system.

Outreach and Mentoring Students

President, Upsilon Pi Epsilon (Alpha Chapter of Arizona): organized SCAI-wide mentorship program that was available to ∼450 Ph.D. students across four doctoral programs. Out of that, ∼100 Ph.D. students were matched as mentors and mentees through the program.

VP External Affairs and Director of Programming Competition, Women in Computer Science (WiCS), ASU: organized a programming competition for ∼100 students; led the GHC 2023 trip selection (∼16 of 200 applicants).

Mentor, ASU DataFest (2023, 2024); travel grant reviewer, Graduate and Professional Student Association (GPSA), ASU.

Panelist: SCAI Ph.D. Students Council, SCAI Ph.D. Open House, SCAI Ph.D. Welcome Event, ISSC-FSE panel on pursuing the Ph.D.

Volunteer, Society of Women Engineers (SWE); Women in Machine Learning (WiML).

Officer, The Golden Key Honors Society (GK).

Sponsorship Director, International Students Society, Multimedia University; **Volunteer**, Information Technology Society, Multimedia University.

Memberships

ACM, IEEE, Women in Machine Learning (WiML), Society of Women Engineers (SWE), Women in Computer Science (WiCS), Women in STEM (WSTEM), Graduate Women's Association (GWA).

Relevant Graduate Coursework

Statistical Machine Learning, Data Mining, Knowledge Representation and Reasoning, Multimedia and Web Databases, Distributed Database Systems, Semantic Web Mining, Data Visualization.

Technical Skills

Languages	Python, C, C++, Java, SQL, Bash, R
ML / DL	PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Keras, OpenCV
Data & Systems	MongoDB, Hadoop MapReduce, Apache Spark, SLURM, Git, Linux/Unix

Press Articles and Media Recognition

May, 2024 **Fahim Tasneema Azad: The 2024 Pitchfork Awards**, *Sun Devil Stories*, ASU Educational Outreach and Student Services; named winner of the 2024 Outstanding Graduate Student Leader award (link).

December, 2023 **Fahim Tasneema Azad: Impact Award, Fall 2023**, *ASU Engineering News*, Ira A. Fulton Schools of Engineering, Arizona State University; profile recognizing the Fall 2023 Fulton Schools Impact Award (link).